Operation Brilliance:
Lighting up Supply Chain Visibility

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Industry is challenged to share serialized shipment *content* data

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What kind of supply chain value can be unlocked?

Organized Retail Crime

Claims Compliance

Authenticity, Certifications, Regulatory…

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The Union Pacific Railroad tracks in Los Angeles in January 2022, littered with debris left by a rash of package thefts from trains. Michael Christopher Brown for The New York Times

https://www.nytimes.com/2024/01/23/magazine/train-robbery-amazon-packages.html

- NRF cited a 93% increase in organized retail theft between 2019 and 2023, with $121 BB in losses in 2023 alone
- Rail theft for class 1 US carriers increased 40% in 2024, over $100 million.

Source: *https://gorail.org/intermodal/strengthening-supply-chains-corca-explained*

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NRF comments to US Department of Transportation

NRF comments regarding protecting America's supply chain from cargo theft:

- Organized Retail Crime (ORC) focus (distinction from petty theft)
  - Increase in violence & volume, polycriminals
- Priority: a.)rail, b.)rail-truck terminals, c.)trucking
- Need for timely & systematic reporting
- Some items are repackaged, impacting product safety
- Insurance stakeholders
- Information access by criminal insiders
- International and regional cargo security councils

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Recent Industry Meeting Notes

- Vertically-integrated retailer custody tracking advantage
- EPC match (serialized product identifiers) to identify rightful owner
- Sensitive data design considerations:
  - Gov. list of “Found” EPCs (limited access)
    - Use for reclaiming stolen goods
  - Gov. list of “Reported Stolen” EPCs (kept confidential/internal to gov agency)
    - Brands/retailers report known stolen items
    - Used by Gov in investigations
- Potential to add product attribute data for advanced AI trend analysis

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November discussion notes:

- Varying products will have varying custody arrangements
  - E.g., some brands retain ownership until DC delivery, others not so
  - Location of logistics unit relevant to clarifying ownership
- Note on CBP thresholds on warnings
- Address accounting challenge when allocating/writing off stolen goods to specific stores
- EPC-level data considered helpful in reducing recovery red tape
- ORC events often follow demand signals

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Store-based example: RFID-based ORC detection

Senate Judiciary Full Committee Hearing on Criminal Networks and Organized Theft

7/15/2025 Testimony excerpt:

"Consider a recent case where Homeland Security Investigations (HSI), working alongside state and local law enforcement, successfully interdicted a consolidation and exfiltration scheme. Stolen apparel was being prepared for packing onto sea-land containers destined to cross the Texas border into Mexico. With the assistance of corporate field investigators, HSI recovered ~2,000 pairs of American Eagle jeans with a retail value exceeding $100,000, along with nearly $2 million in other retailer merchandise. RFID research revealed these stolen AEO items originated from 35 American Eagle stores across 13 states. Evidence from this seizure has already led to the discovery of a second arm of the same criminal group in Utah and has generated additional investigative leads across multiple states. Without the national reach of HSI, we would not have achieved such success in developing this ongoing case; a case far beyond the reach or understanding of any one local or state law enforcement agency given the magnitude of the crimes across a multitude of jurisdictions.

- Mr. Scott McBride, Chief Global Asset Protection Officer, American Eagle Outfitters Inc. (see full testimony link)

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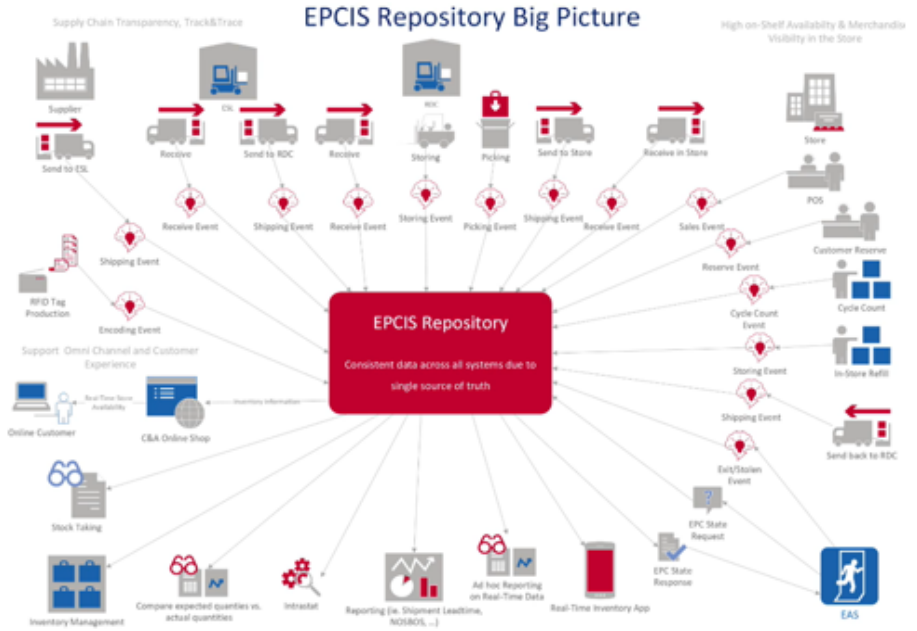
Why hasn't this supply-chain value been unlocked? *(beyond a few vertically-integrated retailers)*

Chicken and the egg: A solution requires a network of EPCIS instances *(the vertically-integrated don't have this barrier)*

A single company's investment requires coordinated trade partner investment

Vertically-Integrated Retailer Example



*"The EPCIS repository is one of the core elements in our RFID IT architecture. Having consistent **data across the whole supply chain** and serving as a single source of truth."*

(See appendix for reference)

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What will unlock this challenge?

Stakeholders, including transportation and logistics providers, need to know:

- where to find trade partner data
- in a manner that is easy, open, and scalable
- and in a manner that benefits all participants

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What kind of data would this serve *to the handler*?

Expected content (serialized)

- claims compliance, ORC detection, traceability, authenticity…

Product attributes

- for operational efficiencies: dimensions, storage conditions…

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What kind of data would this serve *to the sender*?

Waypoint visibility:

- Much greater insight into product movement

Content visibility:

- Last-seen content useful for claims, ORC, authenticity, grey market…

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Operation Brilliance

Lighting up supply chain visibility…

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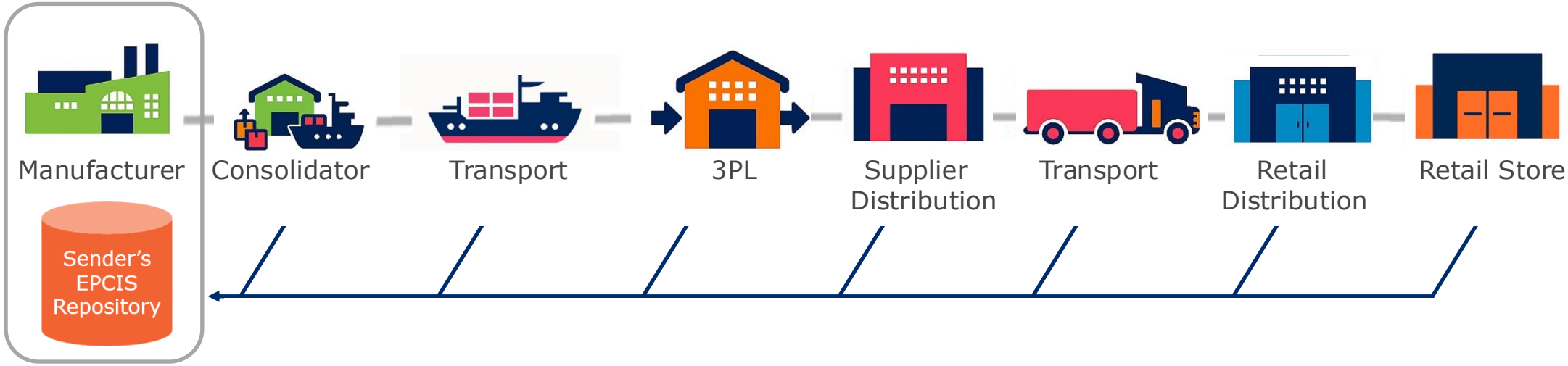
Operation Brilliance:
Easy access to serialized shipment data

- New Tag Data Standard (TDS 2.3)
  - Encoding sender's domain into logistics label removes need for downstream EPCIS instances for many transportation and logistics use cases

- Supply Chain Value:
  - Combatting organized retail crime, claims compliance, product authenticity…



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Brilliant: Web-Enabled Logistics Units

- Data carrier that can "phone home" via interoperable standards
  - Logistics unit handler constructs call to sender's web service
  - Transmits visibility data, receives operational data as needed (content, dimensions…)

- Shipment Label with QR code and RFID data carrier
  - Data carrier data: SSCC (logistics identifier) value with sender's domain
  - Reader-configured data: GLN (Global Location Identifier)
  - Standard LinkTypes: EPCIS & GS1 Digital Link commands

- https://senders_domain/00/<SSCC_Value>?414=<reader_GLN>&linkType=gs1:epcis

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Brilliant: Proof-of-Concept Invitation

- Phase 1: Paper Pilot - data flow & process review
  - Virtual meetings, simulated data and infrastructure
  - Factory scan-pack & shipment data with downstream read points
  - ~8-week engagement Q1 2026 (~3 hours/week per stakeholder)
  - Output report to note legacy systems readiness to leverage data

- Phase 2 scope – physical proof of concept (Q2/Q3)
  - Live production data & reader infrastructure (GS1 US provided EPCIS instance)
  - Requirements / effort to be clarified per phase 1

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Next Steps

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Next Steps (for 12/9 discussion group)

- Design & POC “Brilliant” means of quickly reporting cargo theft
  - Consider/evaluate complementary technologies
- Network & be ready to engage more robust federal resources
  - “Brilliant” POC as springboard for national implementation
- Learn about product category distinctions, especially high-risk to consumers
  - Research means to leverage product attribute data for AI trend analysis
- Develop 2 Workstreams: Supply Chain and Store-Level reporting/response

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Communication Details

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Digital-Link SSCC Data Flow: Content Lookup

Point of Interaction (PoI)

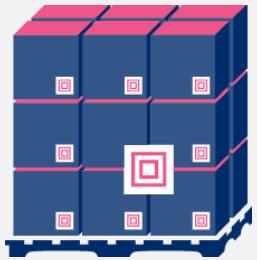
SSCC GS1 Digital Link with EPCIS Link Type

https://senders_domain/00/<sscc>?414=<location>&linkType=gs1:epcis

EPCIS Query and Token

https://epics.example.com/events?MATCH_parentID=<sscc digital link>&EQ_bizStep=packing

Token



Local Application
(not necessarily EPCIS)

JSON Content Data (SGTINs)

Sender

Sender Resolver Redirect

(Web Server interprets and posts "Observe" Event)

Events Endpoint and Credentials

Event Request

EPCIS Event

Sender's EPCIS Repository

1.) Observe Event
2.) Return Data

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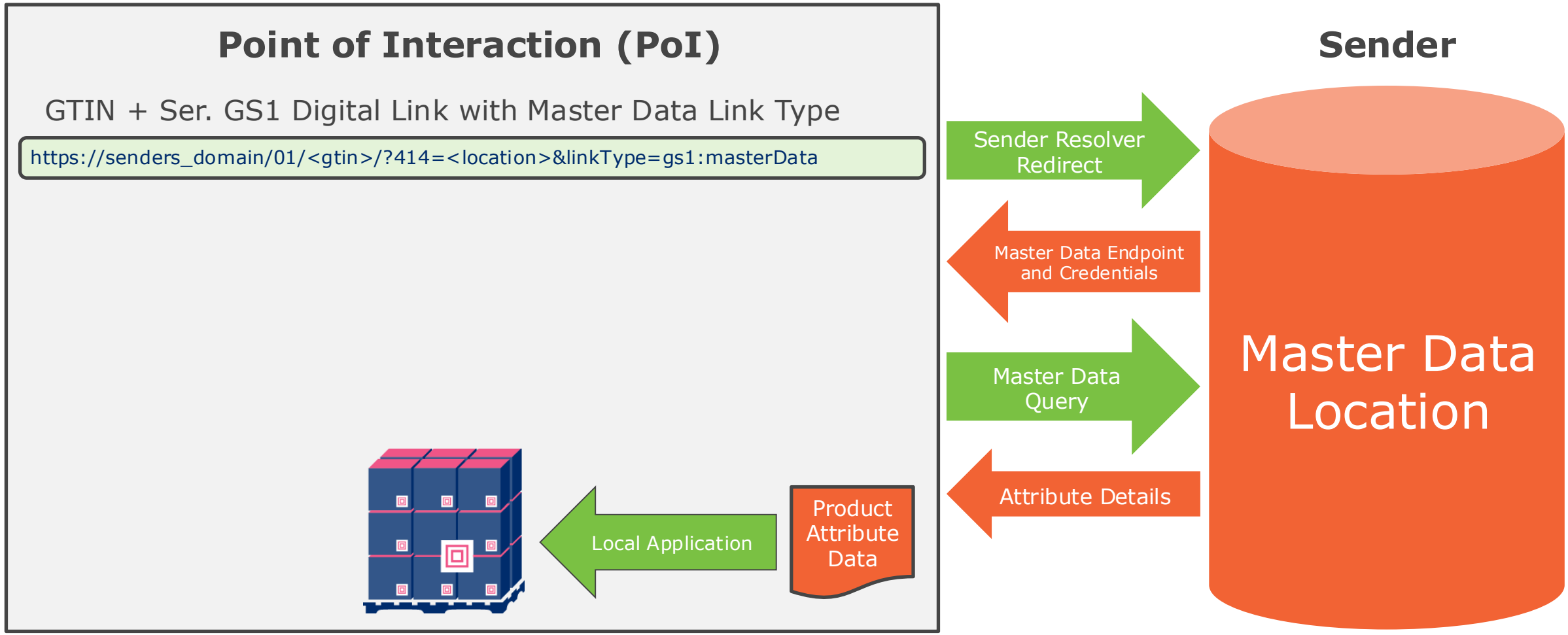
Digital Link SGTIN: Product Attributes



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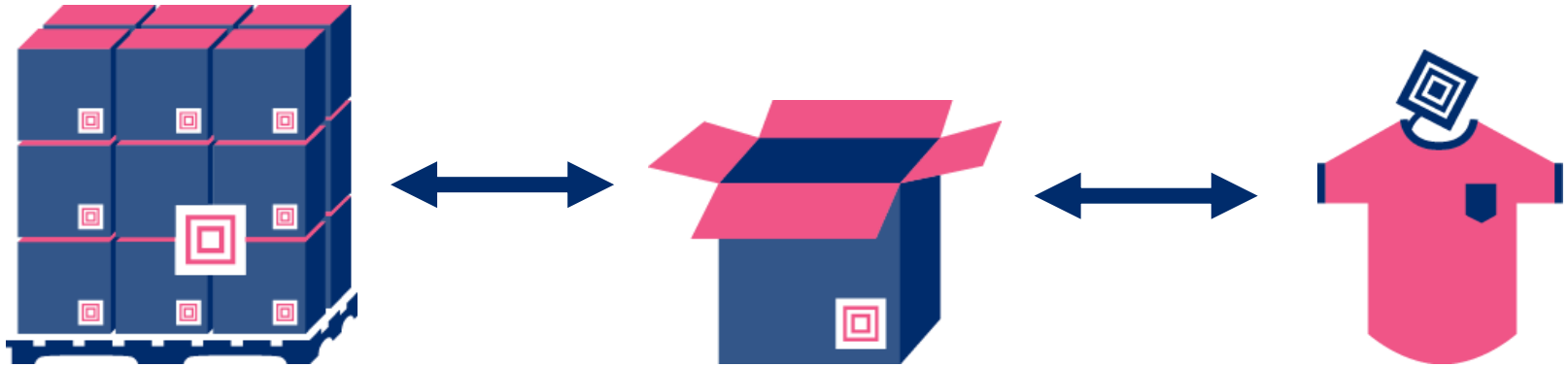
EPCIS Content Lookup Commands (abridged)

Example	Key Identifier	Append Command
Pallet-to-Case	SSCC to content	MATCH_parentID=<SSCC>&EQ_bizStep=packing
Case-to-Pallet	SGTIN to parent	MATCH_epc=<SGTIN>&EQ_bizStep=packing
Case-to-Item	SGTIN to content	MATCH_parentID=<SGTIN>&EQ_bizStep=packing
Purchase Order Info	PO full content	EQ_bizTransaction_po=<PO>&EQ_bizStep=packing



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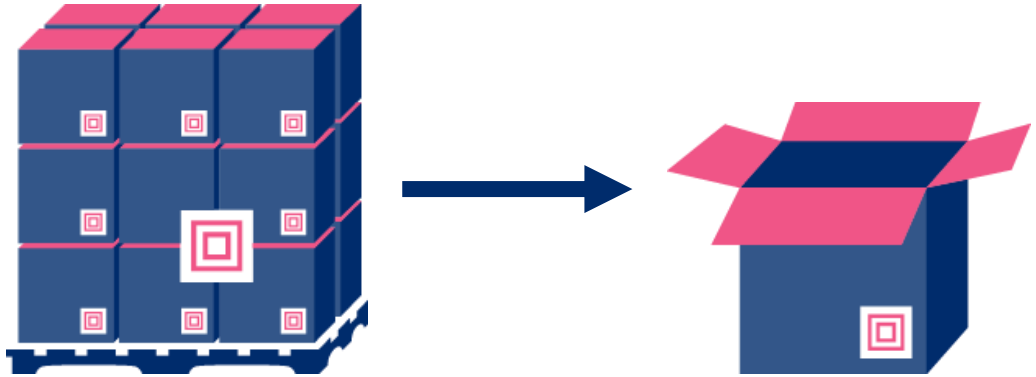
Pallet to Case Packing Details, Serialized Cases

- EPCIS Query for Case to Pallet Aggregation Event
- EPCIS Query Parameters and Values
  - MATCH_parentID = Pallet SSCC Digital Link
  - EQ_bizStep = packing
- Example EPCIS events endpoint URL query:
  - https://epics.example.com/events?MATCH_parentID=<sscc digital link>&EQ_bizStep=packing
- Returned event contains childEPC list with SGTIN Digital Link values of cases



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Case to Pallet Details, Serialized Cases

- EPCIS Query for Case to Pallet Aggregation Event, starting with case SGTIN
- EPCIS Query Parameters and Values
  - MATCH_epc = SGTIN Digital Link
  - EQ_bizStep = packing
- Example EPCIS events endpoint URL query:
  - https://epics.example.com/events?MATCH_epc=<SGTIN digital link>&EQ_bizStep=packing
- Returned event contains pack event when SGTIN was a child and what SSCC(s) it has been packed in

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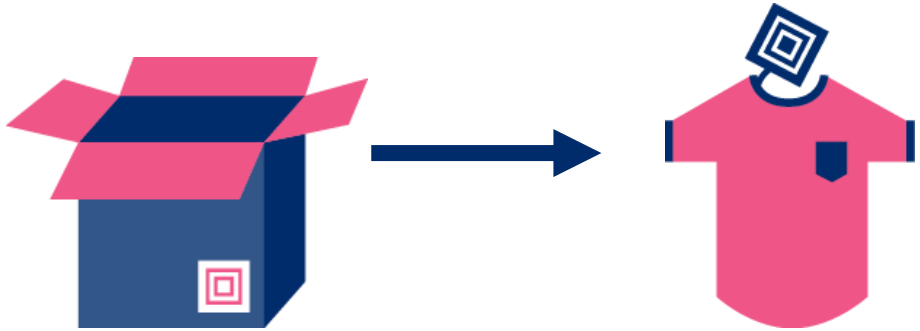
Case to Item Packing Details, Serialized Items

- EPCIS Query for Item to Case Aggregation Event
- EPCIS Query Parameters and Values
  - MATCH_parentID = Case SGTIN Digital Link
  - EQ_bizStep = packing
- Example EPCIS events endpoint URL query:
  - https://epics.example.com/events?MATCH_parentID=<sgtin digital link>&EQ_bizStep=packing
- Returned event contains childEPC list with SGTIN Digital Link values of items



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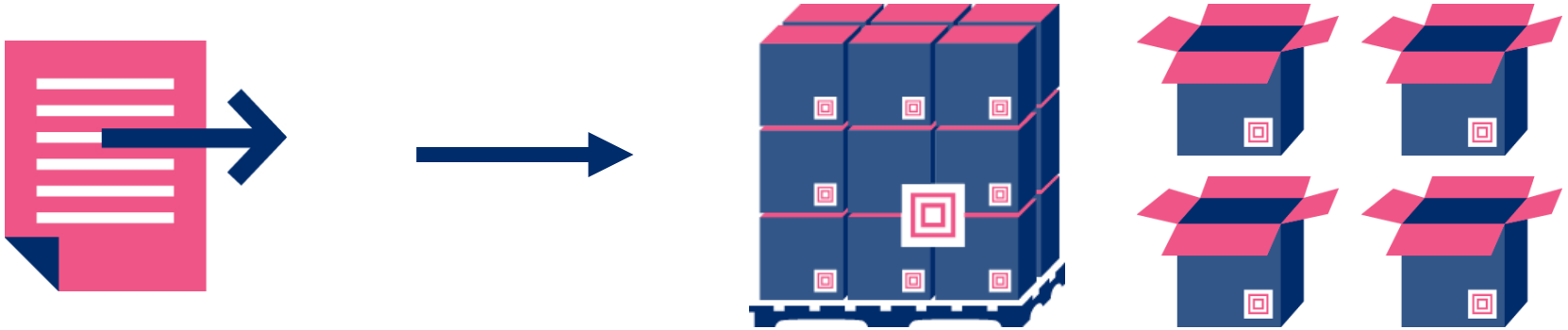
Purchase Order Events

- EPCIS Query for serialized items involved in a specific order
- EPCIS Query Parameters and Values
  - EQ_bizTransaction_po = URI-based PO identifier
  - EQ_bizStep = packing
- Example EPCIS events endpoint URL query:
  - https://epics.example.com/events?EQ_bizTransaction_po =<PO URI>&EQ_bizStep=packing
- Returned event(s) contain all SGTINs and SSCCs involved in order fulfillment
- Requires PO reference in packing event(s)



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Standards Resource Links

- TDS 2.3 Standard (see section 14.6.2.3, page 246) and Artefacts
  - Domain name bit calculator per section 14.5.16
- EPCIS resource landing page
- Note: Update of Tag Data Translation files and encoding utility from 2.2 to 2.3 is TBD
- GS1 Digital Link overview and product narrative

GS1 US stands ready to support a pilot project

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Appendix

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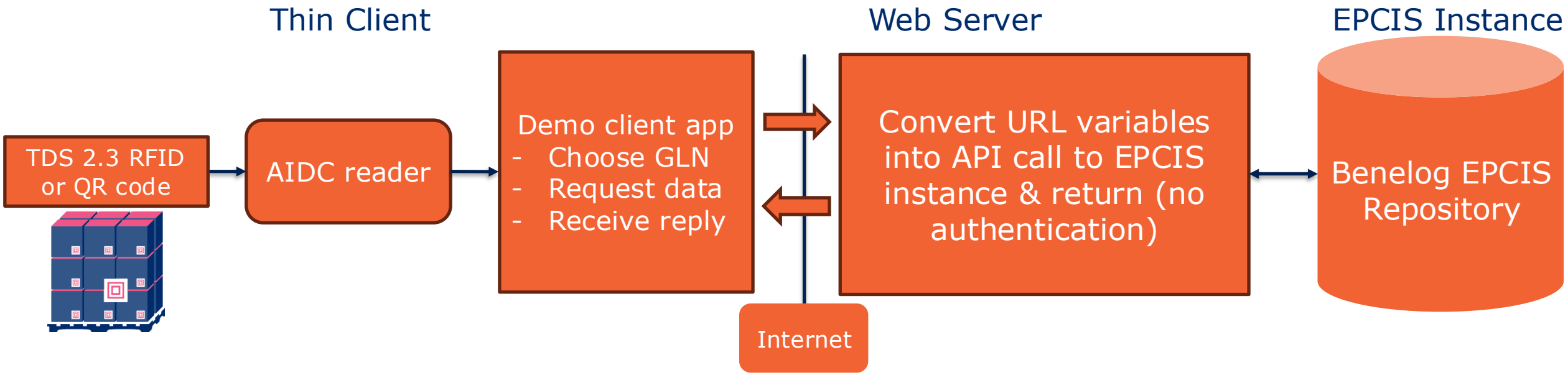
Basic Solution Components (Draft, Phase 2)



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Thin Client Pseudocode

Core requirement for demonstration (e.g., at the Connect conference in June)

- Read case tag (SSCC++)
- Concatenate data, add GLN (Constant), notify source EPCIS instance

Additional “really-nice-to-have” feature:

- After performing the above…
  - Call source EPCIS instance requesting serialized content
  - Read case content tags & compare to expected content
  - Signal case success/fail with a light or buzzer

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Translating a U.P.C. to an EPC

What product (GTIN/U.P.C.)	What specific item	Unique EPC
U.P.C. Company Prefix Item Reference	1732050807 Unique Serial Number	urn:epc:id:sgtin: 0614141.099999 .1732050807

For an overview of RFID, see this [supplier orientation](#) series

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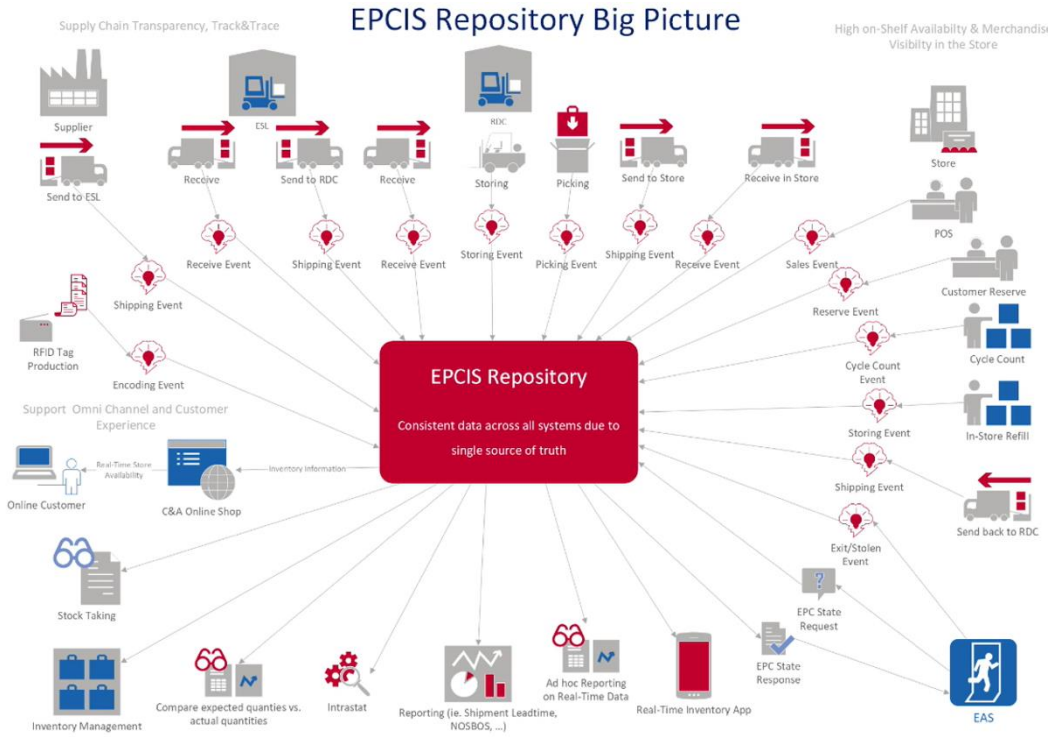
May 11, 2023, C&A Presentation at RFID Journal Live!

C&A

- Vertically integrated multinational retail clothing store
- 1,300 stores in Europe, 300 in Brazil
- $5.5 billion annual sales
- EPCIS repository central to enterprise architecture (centralized choreography)
- Recorded presentation (time index 13:55)



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